

Physics 11 Unit Review:

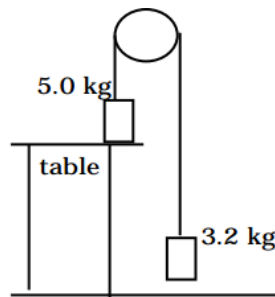
Forces

Name _____

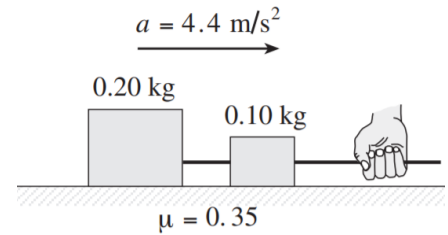
Date _____ Block ____

1.
 - a) How much force (thrust) is needed to accelerate a 2.0 kg block of wood at 4.0 m/s^2 against a 10 N force of friction?
 - b) What is the frictional force due to air resistance on a 0.50 kg object falling vertically with an acceleration of 8.5 m/s^2 ?
 - c) A net force of 0.049 N acts on a body at rest and causes it to acquire a speed of 14.7 m/s in 6.0 s. What is the mass of the body?
 - d) A net force of 450 N acts on an object at rest that has a mass of 120 kg. Calculate distance travelled in 5.0 s.
 - e) A 24 kg chair at rest reaches a velocity of 4.0 m/s in 6.0 s. What was the net force acting on the object?
 - f) A car changes its velocity from 15 m/s to 33 m/s in 2.5 s. If the net force acting on the car is 4500 N, what is the mass of the car?
 - g) A 0.50 kg object accelerates from 21 m/s to 46 m/s in 0.70 s. Calculate the net force acting on the object.
 - h) A 2.0 kg sponge is dropped from rest and is pulled down by gravity. If a 5.0 N force of air resistance acts on sponge, how fast will it be travelling after 6.0 s?
2. Two children are floating side-by-side in space (no friction). 65 kg child pushes 95 kg child with a force of 360 N. Calculate acceleration of each child. Does it matter who pushed whom? Explain.
3. A 5 kg box is accelerated along a level surface at 4.00 m/s^2 by an applied force of 30 N.
Find:
 - a) net force?
 - b) friction force?
 - c) coefficient of friction between block and surface?
4. A 1300 kg vehicle was travelling 60 m/s along a surface with a coefficient of friction of 0.65 when the driver hit the brakes and the car skidded to a stop over a distance of 325 m.
 - a) Find the magnitude of the deceleration of the car
 - b) Find the thrust force that braking action needed to overcome
5.
 - a) A force of 23 N was applied a material with a spring constant of 45 N/m. Object was compressed from starting position of 75 cm. Find final position after compression.
 - b) A force of 14 N was applied a material with a spring constant of 63 N/m. Object was stretched from starting position of 25 cm. Find final position after stretch.

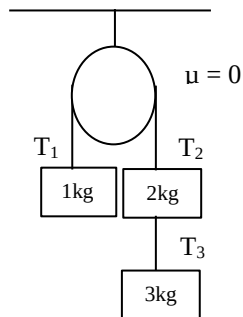
6. System shown is at rest.
Find normal force exerted by table



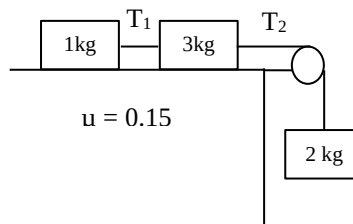
7. System of blocks is being accelerated as shown.
Find:
a) pulling force applied by hand
b) tension in cord connecting masses



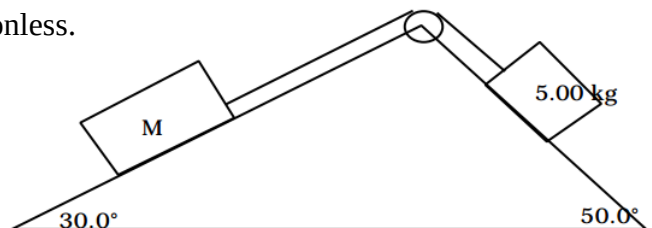
8. Find acceleration
& cord tensions
in scenario shown



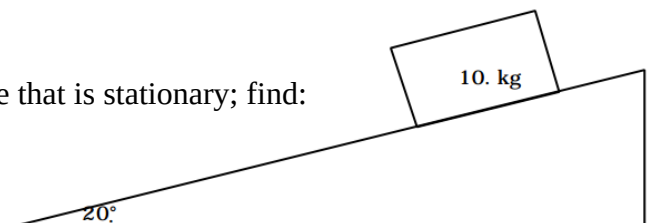
9. Find acceleration & cord tensions
in scenario shown



10. System shown is at rest & surfaces are frictionless.
Find mass M.



11. Consider a 10 kg mass on a 20° incline plane that is stationary; find:
a) normal force
b) friction force
c) minimum coefficient of friction (μ)



12. For system shown, find:

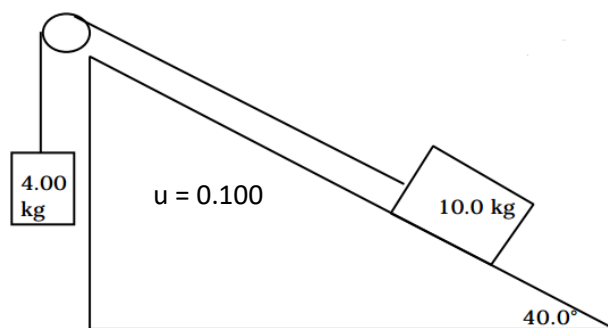
- acceleration (need direction of friction)
- cord tension

13. Revise system shown by:

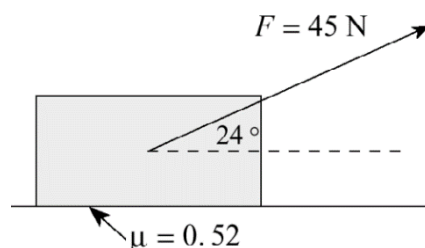
- switching masses
- using acceleration of 4.53 m/s^2 toward 10.0 kg

To find the following values (different than #12):

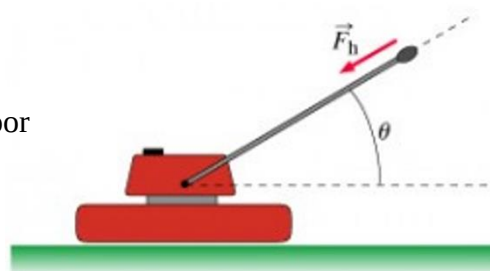
- coefficient of friction
- cord tension



14. a) Student drags a 7.0 kg carton across floor by exerting 45 N as shown. Coefficient of friction between carton & floor = 0.52 . Find acceleration of carton.



- b) 42 kg power mop whose handle is being pushed with an $F_h = 165 \text{ N}$ at an angle of $\theta = 35^\circ$ accelerates at 0.95 m/s^2 . Find coefficient of friction between power mop & floor



KEY

- 18 N
 - 0.65 N
 - 0.020 kg
 - 47 m
 - 16 N
 - 625 kg
 - 18 N
 - 44 m/s
- (65 kg child) $a = 5.5 \text{ m/s}^2$... (95 kg child) $a = 3.8 \text{ m/s}^2$... no matter who pushes action=reaction= 360 N
- 20 N with F_A
 - 10 N opposite F_A
 - 0.2
- 5.5 m/s^2 [backward]
 - 1100 N
- 24 cm
 - 47 cm
- 18 N
- 2.3 N
 - 1.6 N
- 6.5 m/s^2 towards 3 kg ; 16 N ; 10 N
- 2.3 m/s^2 towards 2 kg ; 3.8 N ; 15 N
- 7.66 kg
- 92 N
 - 34 N up ramp
 - 0.36
- 1.16 m/s^2 down ramp
 - 43.9 N
- 0.312
 - 52.7 N
- 2.1 m/s^2
 - 0.19